Regularisation Paths in

Logistic Regression

Watching Features Disappear

A Machine Learning Tutorial

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1. Introduction

Under a logistic regression model, where feature numbering is not limited, and the regression model is not limited by any parameter, any input can be arbitrarily assigned very large coefficients in the model. This is what leads to the unstable and impossible-to-interpret predictions and models used to forecast. This can be handled through regularisation; this adds a penalty term to the loss function, which penalizes large coefficients, so that only the factors which are actually important need to be payed attention to.

This tutorial discusses the two most popular regularisation algorithms, L1 (Lasso) and L2 (Ridge) in a visually unique manner: coefficient path plots. Much can be learnt about how regularisation works, when it chooses the right features, and how to choose the right penalty by sweeping the penalty strength between weak and strong, and monitoring the behaviour of each coefficient. We use the Credit Card Default data, where the question of practical interest is: what are the best characteristics of clients that predict the default on the next payment?

Learning Objectives: (1) explain what regularisation of model coefficients does to the L1 and the L2. (2) Read writing coefficient path plot, identify feature survival. (3) Find out why L1 will shrink and L2 will simply decrease in size. Cross-validation (4) Choose a level of regularisation.

2. Background: Two Ways to Penalize

Both L1 and L2 reg regularisation will contribute a penalty to the logistic regression loss. L2 (Ridge) = squared coefficient C and squared term Sb 2. The magnitude of the absolute obtained values which are punished by L1 (Lasso) = Loss + (1/C) x S|b|. The smaller the penalty, the smaller C - in the scikit-learn notation.

The intuition of geometry is important. Note this as a limiting area in coefficient space. Limit L2 is a circle (or a high dimensional hypersphere) and thus, the optimum solution slides on the surface of the circle - the coefficients gradually diminish, but they never vanish. The L1 limitation is a diamond (or cross-polytope), the axes angles are corners. This optimum is commonly at a corner, i.e. one or more coefficients are all equal to zero. This is the reason that L1 will select the feature automatically.

3. Experimental Setup

The data that we possess is the Credit card default data which encompasses the fact about the client which we define with the factors of the credit history, demographics and utility of money. Binary target means that a client is defaulting. After encoding the categoricals and normalising all the features with StandardScaler (this is necessary to be fair to the features), we split our data 80/20 stratified (i.e. not biased towards a feature due to its scale) with random state=42.

We graph 60 values of C between 10-3 and 103 on a logarithmic scale and then regress BOTH L1 and L2 using SAGA solver. The result is a matrix of coefficient between the penalty strengths, which we plot as coefficient path plots.

4. Results and Analysis

4.1 The Unregularised Baseline

The coefficients of an unregularised logistic regression appear in their magnitude in figure 1. The model approximates non-zero coefficients on the features (and no punishment), one of which can be outstandingly large. The blue bars are the factors that add to the default risk (positive coefficients); the orange bars are the factors that add to the protection (negative coefficients).

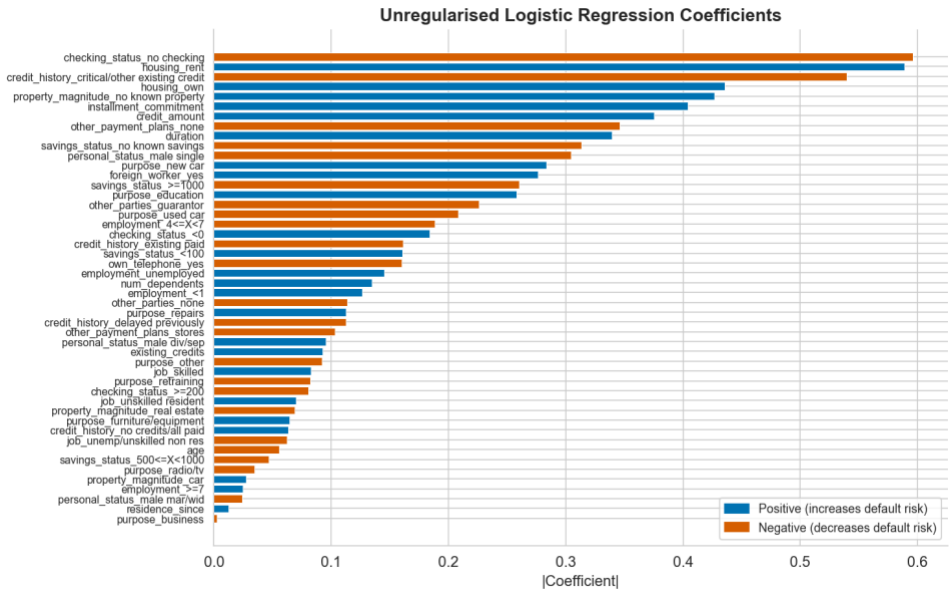

Figure 1: Coefficient magnitudes from unregularised logistic regression. All features receive non-zero weights.

4.2 The L1 (Lasso) Coefficient Path

The figure 2 in this tutorial is the most significant; L1 coefficient path. Delayed backward (weak to strong penalty). The coefficients are nearly non-zero and the features are therefore observed mostly on the right side. As we approach the left, and the penalty is increased, feature coefficients attain a flattened level of coefficient of zero. It possesses 8 best features which have been highlighted by different colours and line styles in order to get them.

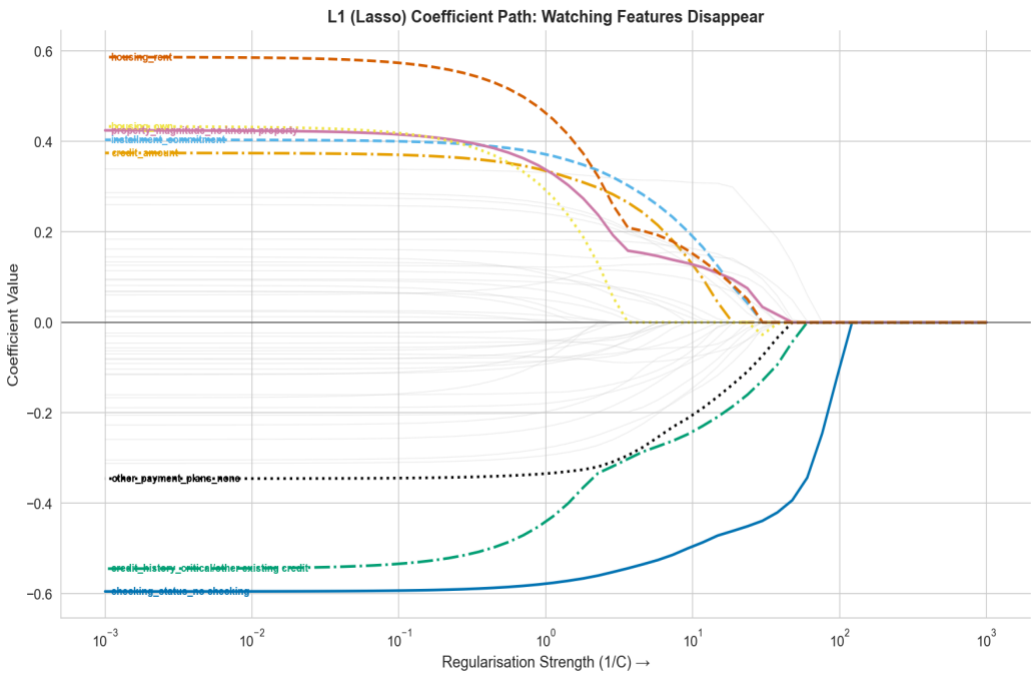

Figure 2: L1 (Lasso) coefficient path — coefficients are driven to exactly zero as penalty increases. Labelled features are the most robust predictors.

Worth mentioning fact: the most strongly important predictors are those which cannot be L1 penified. This gives an unbiased prioritisation of the importance of features of the regularisation behaviour of the model - not a post facto measure. Attributes that tend to fall off fast with high penalty are not predictive in any way, can be safely discarded.

4.3 The L2 (Ridge) Coefficient Path

Same sweep with L2 penalty is provided in figure 3. This difference of L1 can be also observed: all the coefficients themselves are moving towards the decrease to zero simultaneously with the increase in the penalty, but in L1 not a single coefficient tends to decrease to zero. All features are non zero even when they are penalised to the extreme. This is because the gradient of L2 with respect to b (2b) approaches zero at least when the b approaches zero. L1 gradient of a penalty is a constant (+-1) and would only move all directions to the zero point.

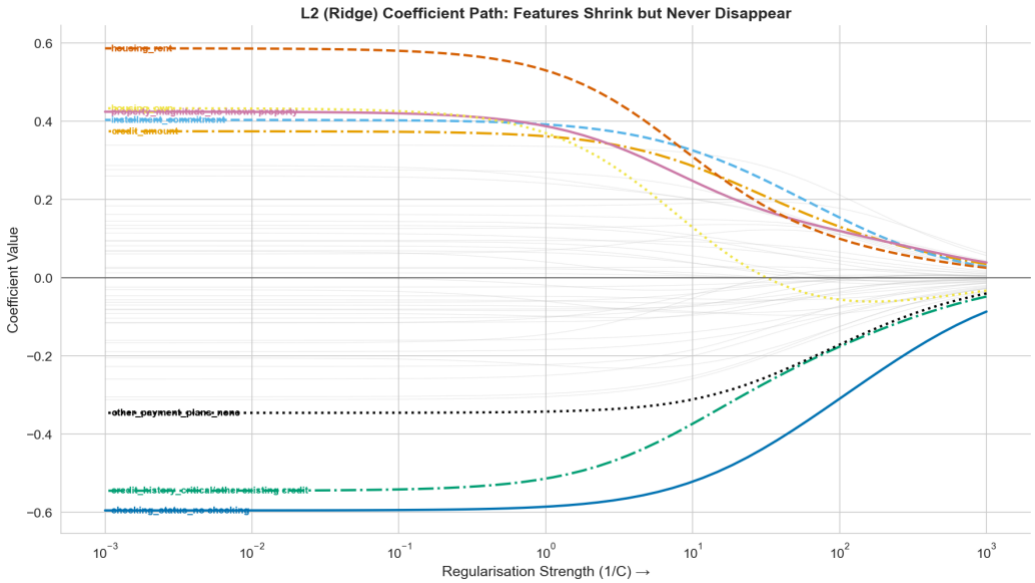

Figure 3: L2 (Ridge) coefficient path — coefficients shrink continuously but never reach zero. No feature selection occurs.

4.4 Accuracy vs Regularisation Strength

Figure 4 and Figure 5 provide a comparison of 5-fold cross-validated stability of L1 and L2 at the full range of penalty strength. The two methods share a characteristic curve, which is too steep (right) in its overshoot and too tight (left), which chokes coefficients and contains a minimum between two. The best C value obtained in each of the methods is marked by the full circle markers.

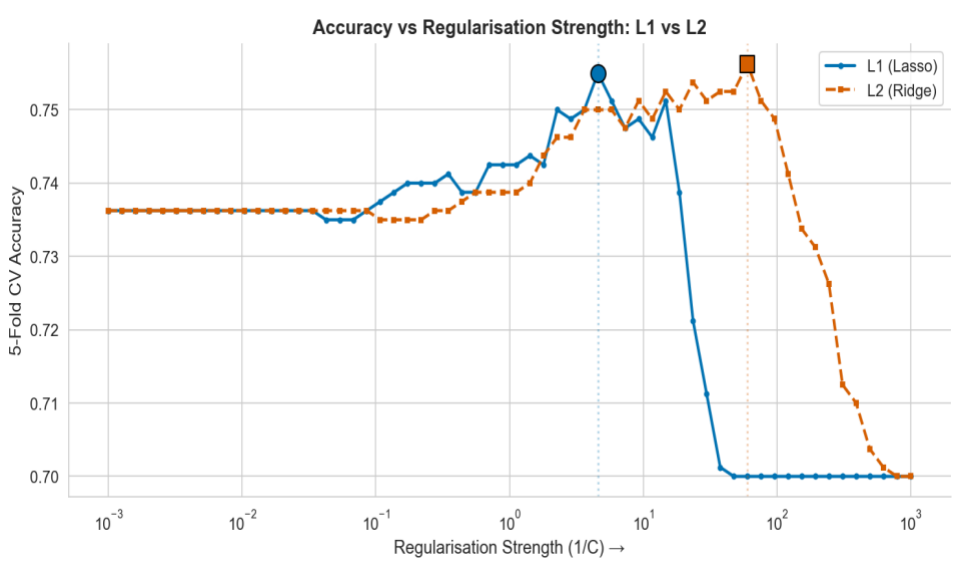

Figure 4: Cross-validated accuracy for L1 and L2 across penalty strengths. Filled markers show optimal C for each method.

It is also notable that L1 and L2 are quite similar in the term of peak accuracy and that implies that the simpler L1 model (only few active features) is equally efficient as the L2 model that

does not ignore any features in this dataset. Such is a pragmatic win: L1 will provide you with such a similar level of accuracy, and will be more interpretable.

4.5 The Sweet Spot: How Many Features Do You Need?

Figure 5 presents the data of the L1 accuracy in combination with the number of the survived features (non-zero coefficients) and provided with the help of the dual-axis plot. The more unpleasant the punishment the less it will take away and accuracy never diminishes but may even increase. We are given a golden mean, with the noisiness taken out of the model, and only the informative remaining there.

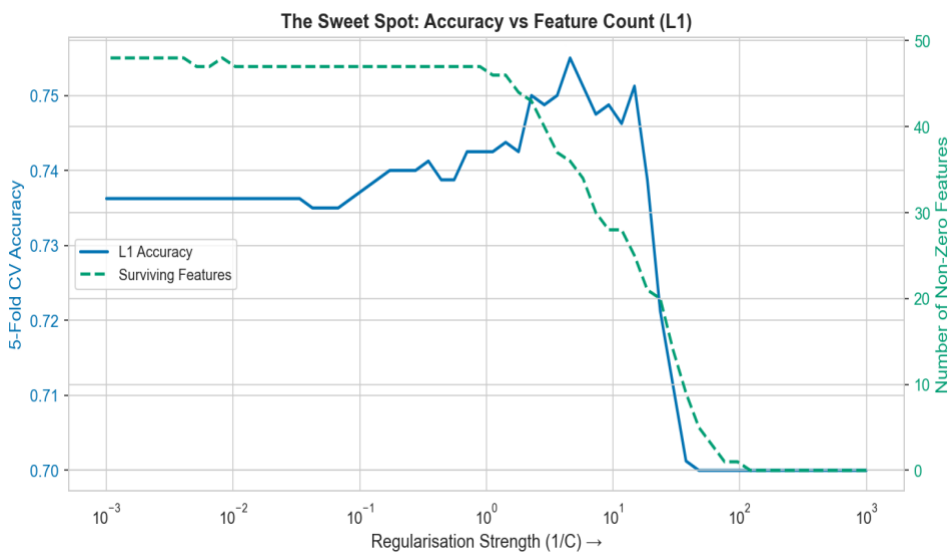

Figure 5: L1 accuracy (blue) vs surviving features (green). The sweet spot shows maximum accuracy with minimal feature count.

The lesson learned here is the relevant two-axis view which gives you all the features you do not require. Regularisation L1 regularisation is in effect a way of preventing overfitting and at the same time performs feature selection which makes the model far simpler to interpret without any drop in performance.

5. When to Use L1 vs L2

Scenario	L1 (Lasso)	L2 (Ridge)
Want feature selection	✓ Eliminates features	✗ Keeps all features
Many irrelevant features	✓ Drops noise	✗ Shrinks but retains
Correlated features	✗ Picks one arbitrarily	✓ Shrinks evenly
Need all features	✗	✓
Interpretability priority	✓ Sparse model	Partial

Table 1: Decision guide for choosing between L1 and L2 regularisation.

6. Limitations and Ethical Considerations

This tutorial has a few flaws. Regularisation is not analogous to the tree based or deep learning models, which have not been tested other than logistic regression. We have used only one set of data, but we could be completely different in other respects. SAGA solver can not reach extreme values of C and this has some small impact on term coefficients. We also did not also investigate Elastic Net (L1+L2 combination) that can be used to address L1 instability in correlated features.

The moral perspective lies in the naturalness of the fact that credit scoring model directly and significantly influences the life of people. Given the fact that such attributes as gender, age, or marital status might not be eliminated by the L1 regularisation, the model could be coded with discriminative trends of the past. The regularisation simplifies the models and reduces their complexity to understand hence facilitates the right to explanation mandated by the regulations such as the EU GDPR. The fake alerts of default will rob individuals of credit and they do not deserve it and this will have far-reaching consequences on their financial wellbeing. Practitioners should make sure that there was no addition of surviving features to features that are secured as proxies.

7. Reproducibility Notes

Both the experiments use stratified splits, random state=42, StandardScaler preprocessing and deterministic settings. The entire code can be accessed by the jupyter notebook that is attached. Instructions Checkout packages Check Requirements.txt Checkout.

8. References

[1] Tibshirani, R. (1996). Regression Shrinkage and Selection via the Lasso. JRSS-B, 58(1), 267-288.

[2] Hoerl, A. E. & Kennard, R. W. (1970). Ridge Regression. Technometrics, 12(1), 55-67.

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[4] Yeh, I.-C. & Lien, C.-H. (2009). Credit card default prediction. Expert Systems with Applications, 36(2).

[5] Scikit-learn documentation: LogisticRegression. https://scikit-learn.org/